

37.1 Poisson regression

Counts are nonnegative integers, their spread grows with their level, and a regressor acts on them multiplicatively. No transformation satisfies all three at once ([Chapter 22](#)): the square root of a Poisson count has nearly constant variance but a mean no longer linear in the regressors, the logarithm a nearly linear mean but a variance that still moves ([Exercise \(B2\)](#)), and neither is defined at zero. The generalized linear model separates the requirements: the link takes the mean, the distribution takes the variance, and nothing is done to the data.

37.1.1 The Poisson as an exponential dispersion family

Write the Poisson mass function in the form [Definition 34.1.1](#) asks for. For $y = 0, 1, 2, \dots$ and $\mu > 0$,

$$\Pr(Y = y) = \frac{e^{-\mu} \mu^y}{y!} = \exp\{y \log \mu - \mu - \log y!\},$$

so the natural parameter is $\theta = \log \mu$, the cumulant function is $b(\theta) = e^\theta$, the dispersion is $\phi = 1$ and the weight is $w = 1$. By [Proposition 34.1.2](#),

$$E(Y) = b'(\theta) = e^\theta = \mu, \quad \text{Var}(Y) = \phi b''(\theta) = e^\theta = \mu,$$

so the variance function is $V(\mu) = \mu$. The Poisson has no free dispersion parameter: once the mean is fixed, so is the variance. That is the most consequential fact in this chapter, and [Sections 37.4](#) and [37.5](#) are about what to do when the data disagree with it. The canonical link $\theta = \log \mu$ also makes the mean multiplicative and keeps it positive at every linear predictor: for once convenience and interpretation point the same way.

Definition 37.1.1 — The Poisson log-linear model

Let $Y_1, \dots, Y_n$ be independent, with Y_i Poisson distributed with mean $\mu_i > 0$, and let $\mathbf{x}_{(i)}$ be the i th row of a model matrix $\mathbf{X}$ of full column rank p . The **Poisson log-linear model** is the generalized linear model ([Definition 34.2.1](#)) with this random component, the log link and the linear predictor $\eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$:

$$\log \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \quad \text{equivalently} \quad \mu_i = \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}) = e^{\beta_1 x_{i1}} \dots e^{\beta_p x_{ip}}. \quad (37.1.1)$$

The second form carries the interpretation. Increasing the j th regressor by one unit, the others held anywhere, multiplies the mean by e^{β_j} , the **rate ratio** of that regressor. It does not depend on the levels of the others, which is the log-scale version of “no interaction”.

Remark (The identity link). Nothing forces the log link. The model $\mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ gives regressors additive effects on the count, which is sometimes the question asked: an intervention that prevents three events a year, not one that cuts events by ten per cent. The price is a constrained parameter space, since $\mathbf{x}_{(i)}^\top \boldsymbol{\beta} > 0$ is needed for every i and the constraint can bind at the maximum ([Exercise \(C1\)](#)).

37.1.2 The likelihood equations

Theorem 37.1.1 — The Poisson likelihood equations

In the model of Definition 37.1.1 the log-likelihood is

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \{y_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta} - \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})\} - \sum_{i=1}^n \log y_i!, \quad (37.1.2)$$

and, writing $\boldsymbol{\mu} = \boldsymbol{\mu}(\boldsymbol{\beta})$ for the vector of means and $\mathbf{W} = \text{diag}(\mu_1, \dots, \mu_n)$:

- (a) the score is $\partial\ell/\partial\boldsymbol{\beta} = \mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu})$, so the likelihood equations are $\mathbf{X}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$;
- (b) the observed and expected information coincide and equal $\mathbf{X}^\top \mathbf{W} \mathbf{X}$;
- (c) ℓ is concave, and strictly concave when $\text{rank}(\mathbf{X}) = p$, so a stationary point is the unique maximum.

Proof. The Poisson has $\phi = 1$, $w_i = 1$, $V(\mu) = \mu$ and, under the canonical link, $\partial\mu_i/\partial\eta_i = \mu_i$. Substituting these in Theorem 34.3.1(d) gives the score $\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu})$ and observed information equal to expected information $\mathbf{X}^\top \text{diag}\{w_i V(\mu_i)\} \mathbf{X} = \mathbf{X}^\top \mathbf{W} \mathbf{X}$, which are (a) and (b). For (c), that Hessian is nonnegative definite because $\mathbf{a}^\top \mathbf{X}^\top \mathbf{W} \mathbf{X} \mathbf{a} = \sum_i \mu_i (\mathbf{x}_{(i)}^\top \mathbf{a})^2 \geq 0$, and the sum vanishes only if $\mathbf{X} \mathbf{a} = \mathbf{0}$, since every $\mu_i > 0$. Under full column rank that forces $\mathbf{a} = \mathbf{0}$, so ℓ is strictly concave. ■

The general score of Theorem 34.3.1 carries a factor $(\partial\mu_i/\partial\eta_i)/V(\mu_i)$, here $\mu_i/\mu_i = 1$; what is left has the form of the normal equations $\mathbf{X}^\top(\mathbf{y} - \hat{\mathbf{Y}}) = \mathbf{0}$, with the fitted mean vector in place of the projection $\mathbf{M}\mathbf{y}$. The residual $\mathbf{y} - \hat{\boldsymbol{\mu}}$ is orthogonal to every column of $\mathbf{X}$, although $\hat{\boldsymbol{\mu}}$ is not a linear function of $\mathbf{y}$ and there is no projection in sight (Exercise (B2)).

Theorem 37.1.2 — Fitted values reproduce observed margins

Let $\hat{\boldsymbol{\mu}}$ solve the likelihood equations of Theorem 37.1.1. Then $\mathbf{x}_j^\top \hat{\boldsymbol{\mu}} = \mathbf{x}_j^\top \mathbf{y}$ for every column $\mathbf{x}_j$ of $\mathbf{X}$, and more generally $\mathbf{a}^\top \hat{\boldsymbol{\mu}} = \mathbf{a}^\top \mathbf{y}$ for every $\mathbf{a} \in \mathcal{C}(\mathbf{X})$. In particular:

- (a) if the model has an intercept, the fitted counts sum to the observed total, $\sum_i \hat{\mu}_i = \sum_i y_i$;
- (b) if $\mathbf{X}$ contains the indicator of a group G , the fitted counts of that group sum to its observed total, $\sum_{i \in G} \hat{\mu}_i = \sum_{i \in G} y_i$;
- (c) if $\mathbf{X}$ contains a quantitative regressor $\mathbf{x}_j$, the fitted counts reproduce the observed $\mathbf{x}_j$ -weighted total.

Proof. The column identities are Corollary 34.3.2 with $w_i = 1$: the likelihood equations say $\mathbf{X}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$, that is, $\mathbf{x}_j^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = 0$ for each j . The extension to $\mathcal{C}(\mathbf{X})$ is linearity: any $\mathbf{a} \in \mathcal{C}(\mathbf{X})$ is $\mathbf{X}\mathbf{c}$ for some $\mathbf{c}$, and $\mathbf{a}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{c}^\top \mathbf{X}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = 0$. Parts (a), (b) and

(c) are this statement for $\mathbf{a} = \mathbf{1}$, for the group indicator, and for $\mathbf{x}_j$.

■

This is what makes log-linear models for contingency tables behave so tidily; [Section 37.3](#) uses it throughout.

37.1.3 When the maximum likelihood estimate exists

Strict concavity says that a maximum, if attained, is unique; not that one is attained. The log-likelihood [Eq. 37.1.2](#) can increase along a ray for ever, and then the fitting algorithm diverges. The condition below is the Poisson counterpart of separation in logistic regression (Theorem 35.3.1).

Proposition 37.1.3 — Existence of the maximum likelihood estimate

Let $\text{rank}(\mathbf{X}) = p$. The log-likelihood [Eq. 37.1.2](#) attains its maximum at a finite $\hat{\boldsymbol{\beta}}$ if and only if there is **no** vector $\mathbf{d} \neq \mathbf{0}$ with

$$\mathbf{x}_{(i)}^\top \mathbf{d} \leq 0 \text{ for every } i, \quad \text{and} \quad \mathbf{x}_{(i)}^\top \mathbf{d} = 0 \text{ whenever } y_i > 0. \quad (37.1.3)$$

Proof. Write $u_i = \mathbf{x}_{(i)}^\top \mathbf{d}$ and, for a fixed $\boldsymbol{\beta}_0$,

$$\ell(\boldsymbol{\beta}_0 + t\mathbf{d}) = \sum_i y_i \eta_i^0 + t \sum_i y_i u_i - \sum_i e^{\eta_i^0} e^{tu_i} - \sum_i \log y_i!,$$

where $\eta_i^0 = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_0$. Suppose such a $\mathbf{d}$ exists. Then $\sum_i y_i u_i = 0$, because every term with $y_i > 0$ has $u_i = 0$. Since every $u_i \leq 0$, each e^{tu_i} is nonincreasing in t , so $\ell(\boldsymbol{\beta}_0 + t\mathbf{d})$ is nondecreasing in t ; and it is strictly increasing unless every $u_i = 0$, which full column rank excludes. So the supremum is approached only as $t \rightarrow \infty$ and is not attained.

Conversely, suppose no such $\mathbf{d}$ exists, and fix a unit vector $\mathbf{d}$. Either some $u_i > 0$, and $\ell(\boldsymbol{\beta}_0 + t\mathbf{d}) \rightarrow -\infty$ because an exponential beats a linear term; or every $u_i \leq 0$, in which case some i with $y_i > 0$ has $u_i < 0$, so $\sum_i y_i u_i < 0$ while $-\sum_i e^{\eta_i^0} e^{tu_i} \leq 0$, and again $\ell \rightarrow -\infty$. So there is a $t_{\mathbf{d}}$ beyond which $\ell(\boldsymbol{\beta}_0 + t\mathbf{d}) < \ell(\boldsymbol{\beta}_0) - 1$. Continuity alone does not make that threshold uniform in $\mathbf{d}$; concavity does. By part (c) the difference quotient $\{\ell(\boldsymbol{\beta}_0 + t\mathbf{d}) - \ell(\boldsymbol{\beta}_0)\}/t$ is nonincreasing in t , so once it falls below $-1/t_{\mathbf{d}}$ it stays below, and by continuity it does so on a neighbourhood of $\mathbf{d}$. Finitely many such neighbourhoods cover the unit sphere, giving a single t^* outside whose ball $\ell < \ell(\boldsymbol{\beta}_0) - 1$; and a continuous function on that closed ball attains its maximum.

■

The condition fails when a regressor is the indicator of a set of observations whose counts are all zero: take $\mathbf{d}$ to be minus that coordinate direction. The coefficient runs to $-\infty$ and the algorithm reports a huge negative number with an enormous standard error. A ridge or lasso

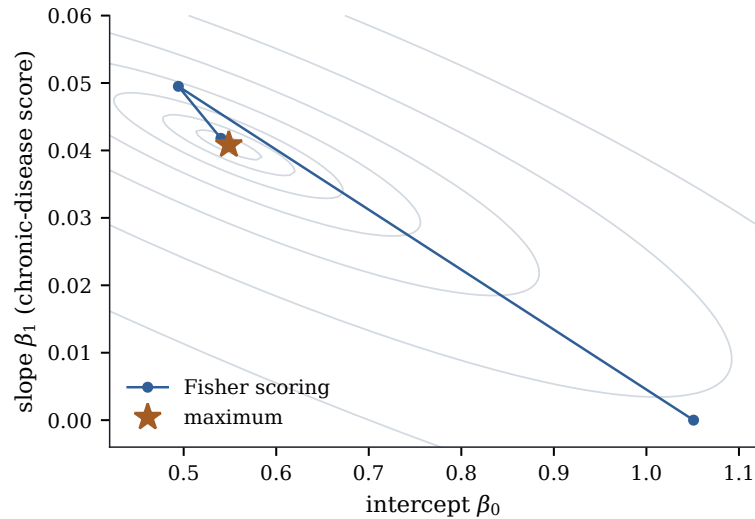


Figure 37.1.1: The log-likelihood of a Poisson log-linear model with an intercept and one regressor, for the physician-visit data used below in this section, with the Fisher scoring path from the start $\beta = (\log \bar{y}, 0)^\top$. Contours are drawn at fixed drops below the maximum.

penalty (Chapter 27, Chapter 30) makes the penalized log-likelihood coercive and restores a finite maximizer.

37.1.4 Fitting

By Theorem 34.4.1, Fisher scoring is iteratively reweighted least squares: at the current β form the working response and weights

$$z_i = \eta_i + \frac{y_i - \mu_i}{\mu_i}, \quad w_i = \mu_i, \quad (37.1.4)$$

and set $\beta \leftarrow (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{z}$. Because the expected and observed information agree (Theorem 37.1.1(b)), Fisher scoring and Newton–Raphson coincide here, and by concavity the iteration converges from any start whenever a maximum exists. The same weighted least squares step appears in Proposition 20.6.1 and Theorem 21.3.1; only the weights differ. Figure 37.1.1 draws the log-likelihood surface of a two-regressor fit with the path from a poor start: the contours are closed and nested, as concavity requires, and the first step already lands near the maximum.

37.1.5 Interpretation and inference

The estimate $\hat{\beta}$ is consistent and asymptotically normal under the conditions of Theorem 34.5.1, with asymptotic covariance $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$ at the estimate. Exponentiating the ends of the Wald interval $\hat{\beta}_j \pm z_{1-\alpha/2} \text{se}(\hat{\beta}_j)$ gives an interval for the rate ratio e^{β_j} , better than a delta method applied to $e^{\hat{\beta}_j}$: the log scale is where the normal approximation is accurate, and the transformed interval stays positive.

Example. (Physician visits in a health insurance experiment)

The RAND Health Insurance Experiment sample distributed with `statsmodels` records 20190 person-years. The response is the number of outpatient physician visits in the year, with mean 2.860 and a variance 7.093 times as large; Example (Physician visits are counts) used the same data to show how badly a normal linear model describes it.

Fit [Eq. 37.1.1](#) with an intercept, the logarithm of one plus the coinsurance rate (`lncoins`), an indicator of an individual-deductible plan (`idp`), a physical-limitation index, a chronic-disease score, and three indicators of self-rated health (good, fair, poor, against excellent). Fisher scoring converges in 7 steps. The coefficient of `lncoins` is -0.0712 , standard error 0.0022: a rate ratio of 0.9313 per unit, Wald interval (0.9272, 0.9354). Moving across its whole range of 4.615, from free care to the highest coinsurance plan, multiplies expected visits by 0.720. The disease score is not a count of diseases but a continuous index running to 58.6; each unit of it multiplies expected visits by 1.0353, so moving from its lower quartile 6.90 to its upper quartile 13.73 multiplies them by 1.267. Poor self-rated health multiplies them by 1.2441 against excellent health. The fitted counts sum to the observed total, and within each health group to that group's total, as [Theorem 37.1.2](#) requires.

```
import numpy as np
import statsmodels.api as sm

data = sm.datasets.randhie.load_pandas().data          # 20190 person-years, public
domain
y = data["mdvis"].to_numpy(float)                     # outpatient physician visits
in the year
names = ["intercept", "lncoins", "idp", "physlm", "disea", "hlthg", "hlthf", "hlthp"
]
X = np.column_stack([np.ones(len(y))] + [data[v].to_numpy(float) for v in names
[1:]])

def poisson_fit(X, y, offset=0.0, tol=1e-11, maxit=50):
    """Fisher scoring for log mu = X beta + offset; returns beta and its iterates.
    """
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(y.mean())
    path = [beta]
    for _ in range(maxit):
        mu = np.exp(X @ beta + offset)                # current fitted means
        z = X @ beta + (y - mu) / mu                  # working response
        XW = X * mu[:, None]                          # weights w_i = mu_i
        step = np.linalg.solve(X.T @ XW, XW.T @ z)
        converged = np.max(np.abs(step - beta)) < tol
        beta = step
        path.append(beta)
    if converged:
        break
    return beta, np.array(path)

beta, path = poisson_fit(X, y)
mu = np.exp(X @ beta)
se = np.sqrt(np.diag(np.linalg.inv(X.T @ (X * mu[:, None]))))
```

```
print("largest entry of X^T (y - mu):", np.abs(X.T @ (y - mu)).max())
for name, b, s in zip(names, beta, se):
    print(f"{name:10s} {b:8.4f} ({s:.4f})    rate ratio {np.exp(b):.4f}")
```

[Run this code online](#)

The deviance of the Poisson fit (Definition 34.5.1) is

$$D = 2 \sum_{i=1}^n \left\{ y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \right\}, \quad (37.1.5)$$

with the convention $0 \log 0 = 0$. If the model has an intercept the second term sums to zero by [Theorem 37.1.2\(a\)](#), and the deviance reduces to $2 \sum_i y_i \log(y_i / \hat{\mu}_i)$, the form usually written G^2 ; because $\phi = 1$ the deviance and the scaled deviance coincide.

Differences of deviances between nested models behave in the usual way: by [Theorem 34.5.5](#) the drop is asymptotically χ^2 on the difference in dimension, and it is the likelihood ratio statistic of [Proposition 34.5.3](#). The deviance of a *single* model is a different matter: its χ^2 calibration needs the fitted means to grow, which happens for grouped data with large cells and not for ungrouped records with small counts. In [Example \(Physician visits in a health insurance experiment\)](#) the deviance is 84569.7 on 20182 degrees of freedom and the Pearson statistic is 127526.5, ratios of 4.190 and 6.319. No formal test is available, but ratios of this size are a loud signal of misfit.

```
dev = 2 * np.sum(np.where(y > 0, y * np.log(np.where(y > 0, y, 1) / mu), 0.0) - (y -
    mu))
pearson = np.sum((y - mu) ** 2 / mu)
df = len(y) - X.shape[1]
print(f"deviance {dev:.1f} on {df} df    (ratio {dev / df:.3f})")
print(f"Pearson {pearson:.1f} on {df} df    (ratio {pearson / df:.3f})")
print("fitted total", mu.sum(), " observed total", y.sum())
```

[Run this code online](#)

Both statistics are sums of squared residuals: the Pearson residual $r_i^P = (y_i - \hat{\mu}_i) / \sqrt{\hat{\mu}_i}$ and the deviance residual $r_i^D = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}$, with d_i the i th term of [Eq. 37.1.5](#). These are the count analogues of [Definition 20.1.1](#); both are badly skewed when $\hat{\mu}_i$ is small ([Exercise \(B3\)](#)), so a normal probability plot of them tells one little. Leverage and influence transfer from [Chapter 20](#) through the weights: the hat matrix of the last weighted least squares step of [Proposition 20.6.1](#) plays the role of **M**, and a one-step Cook's distance ([Definition 20.4.1](#)) built from it is the standard influence measure for a generalized linear model.

Idea. The Poisson log-linear model puts the linear predictor on the log scale, so its coefficients are log rate ratios, and it ties the variance to the mean, so it has no spare parameter for extra variability. The rest of the chapter follows from taking one or other of those decisions seriously. A deviance far larger than its degrees of freedom has three possible causes — a wrong mean model, a wrong variance function, or dependent observations — and reaching for an overdispersion parameter before checking the mean model is a common and expensive mistake, because an inflated variance hides exactly the lack of fit that would have told you what to add.

37.1.6 Exercises

37.1.6.1 A. Check your understanding

Exercise A1.

A Poisson log-linear model for the number of insurance claims has coefficient 0.182 on an indicator of urban residence. State the effect on the expected number of claims in words, and give the multiplicative effect of the indicator on the *odds* that a policy has at least one claim under the same model. Why is the second quantity not simply $e^{0.182}$?

Exercise A2.

Show directly from [Theorem 37.1.1](#) that in the model with an intercept only, the maximum likelihood estimate of the common mean is $\bar{y}$, and that the deviance equals $2 \sum_i y_i \log(y_i/\bar{y})$.

Solution. With $\mathbf{X} = \mathbf{1}$ the likelihood equation is $\sum_i (y_i - e^\beta) = 0$, so $e^\beta = \bar{y}$. Substituting into [Eq. 37.1.5](#) and using $\sum_i (y_i - \bar{y}) = 0$ leaves $2 \sum_i y_i \log(y_i/\bar{y})$.

37.1.6.2 B. Practice

Exercise B1.

Counts y_{ij} , $j = 1, \dots, n_i$, are observed in c groups, and the model is $\log \mu_{ij} = \beta_i$, one parameter per group. Show that $\hat{\mu}_{ij} = \bar{y}_{i\cdot}$, that $\widehat{\text{Var}}(\hat{\beta}_i) = 1/(n_i \bar{y}_{i\cdot})$, and that a Wald interval for μ_h/μ_i is $\exp\{\hat{\beta}_h - \hat{\beta}_i \pm z\sqrt{1/(n_h \bar{y}_{h\cdot}) + 1/(n_i \bar{y}_{i\cdot})}\}$.

Solution. [Theorem 37.1.2\(b\)](#) applied to the group indicators gives $n_i \hat{\mu}_i = \sum_j y_{ij}$. The information $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ is diagonal with entries $n_i \mu_i$, so the estimates are independent with variances $1/(n_i \mu_i)$; the variance of their difference is the sum, and exponentiating a Wald interval for $\beta_h - \beta_i$ gives the stated interval.

Exercise B2.

In the setting of [Exercise \(B1\)](#), show that the likelihood ratio statistic for $H_0 : \mu_1 = \dots = \mu_c$ is $2 \sum_i n_i \bar{y}_{i\cdot} \log(\bar{y}_{i\cdot}/\bar{y})$, where $\bar{y}$ is the overall mean, and state the conditions under which its $\chi^2(c-1)$ calibration is reasonable.

Exercise B3.

For a Poisson fit the Pearson residual is $r_i^P = (y_i - \hat{\mu}_i)/\sqrt{\hat{\mu}_i}$. Show that it is strongly skewed when μ_i is small, and that the Anscombe residual $\frac{3}{2}(y_i^{2/3} - \hat{\mu}_i^{2/3})/\hat{\mu}_i^{1/6}$ has variance about one, by applying the delta method to $Y^{2/3}$.

37.1.6.3 C. Going deeper

Exercise C1.

Consider a Poisson model with the identity link, $\mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$. Show that the log-likelihood is concave on the open set where every $\mu_i > 0$, but that the maximum can occur on its boundary, where some $\hat{\mu}_i = 0$; what happens then to the standard errors an algorithm that ignores the constraint reports? Repeat for the additive *rate* model $\mu_i = t_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ of [Section 37.2](#), whose coefficients are rate *differences*: show that its likelihood equations are $\sum_i t_i (y_i - \mu_i) \mathbf{x}_{(i)} / \mu_i = \mathbf{0}$, and say why an epidemiologist asking how many events an intervention prevents wants it.

Exercise C2.

Give a data set with $n = 4$, $p = 2$ for which the condition of [Proposition 37.1.3](#) fails. Then add a ridge penalty $\lambda \|\boldsymbol{\beta}\|^2$ (Definition 27.2.1) and show that the penalized log-likelihood has a finite maximizer for every $\lambda > 0$.

Solution. Take $\mathbf{X}$ with intercept and a regressor $x = (0, 0, 1, 1)^\top$ and counts $\mathbf{y} = (3, 5, 0, 0)^\top$. Then $\mathbf{d} = (0, -1)^\top$ has $\mathbf{x}_{(i)}^\top \mathbf{d} \leq 0$ everywhere and $= 0$ at the two observations with $y_i > 0$, so the condition fails: the slope runs to $-\infty$ while the intercept converges to $\log 4$. Adding $-\lambda \|\boldsymbol{\beta}\|^2$ makes the objective tend to $-\infty$ in every direction, since the log-likelihood grows at most linearly along any ray on which it grows at all, so a finite maximizer exists; it is unique by strict concavity of the penalized objective.